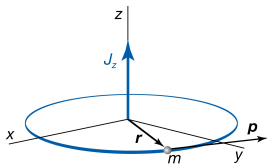
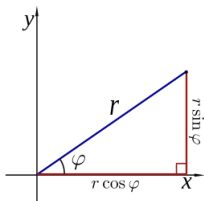
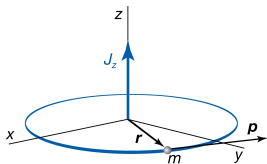


Particle on a ring or particle rotating around a point in a plane freely:

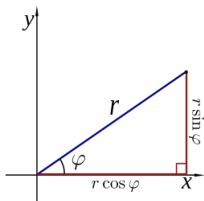
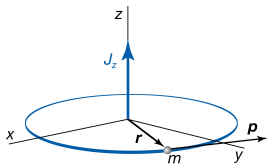
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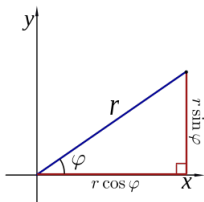
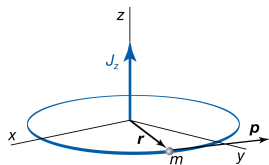


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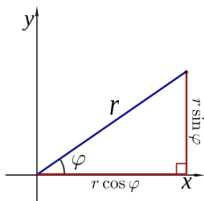
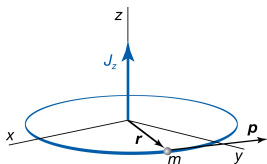
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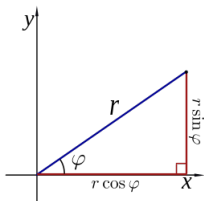
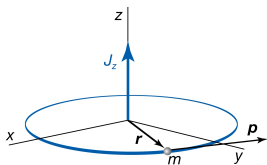
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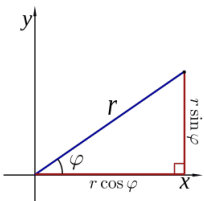
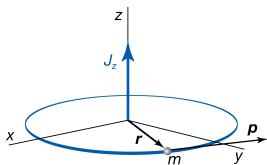
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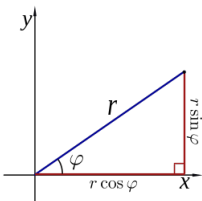
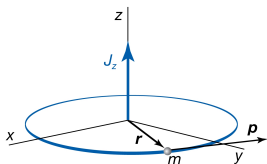
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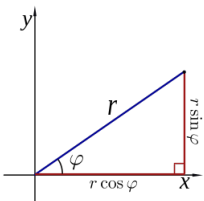
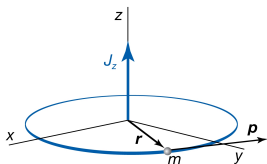
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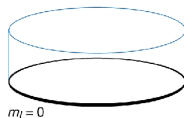
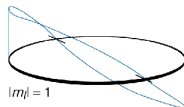
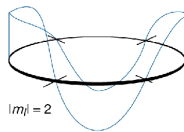
$\therefore m$ is an integer

$$\text{Normalise the function and find } N = \sqrt{\frac{1}{2\pi}}$$

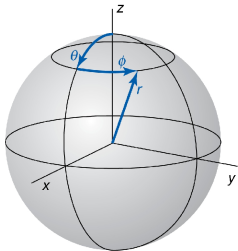
energies are quantised

$$E = \frac{J^2}{2I} = \frac{m^2 \hbar^2}{2I}, \quad m = 0, 1, 2, \dots$$

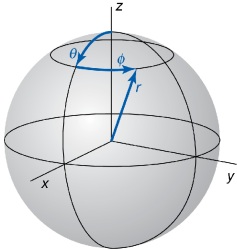
real or imaginary
part of function



3-D rotational motion :
particle on a sphere

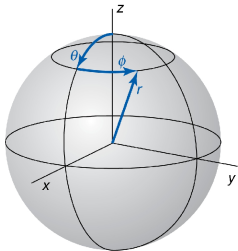


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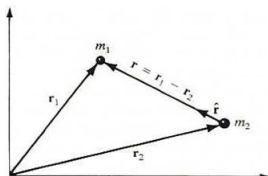


equivalent to a rigid
rotor

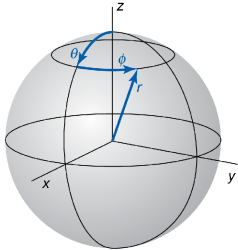
3-D rotational motion :
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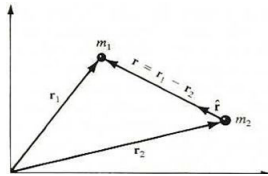
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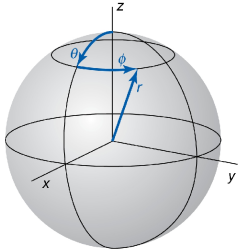


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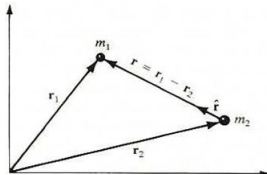


$$\vec{R}_{\text{CM}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{M(=m_1 + m_2)}; \quad \vec{r} = \vec{r}_2 - \vec{r}_1$$

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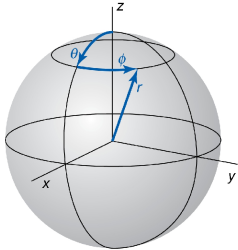
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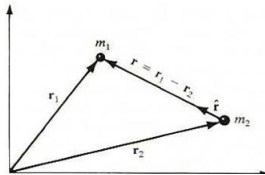
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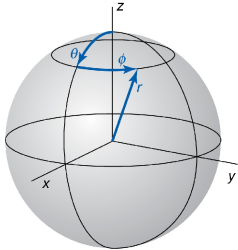


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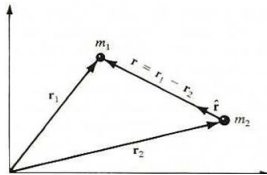
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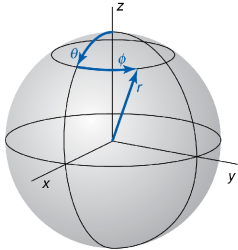
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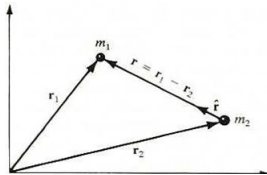
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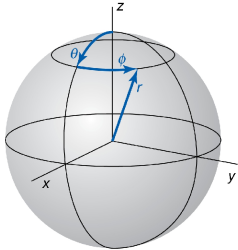
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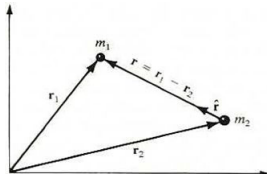
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$$H = \frac{p_{\text{CM}}^2}{2M} + \frac{p^2}{2\mu} + V(\vec{r})$$

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$$\vec{R}_{\text{CM}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{M (=m_1 + m_2)}; \quad \vec{r} = \vec{r}_2 - \vec{r}_1$$

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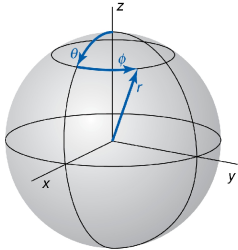
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sit on C.M. and observe relative motion

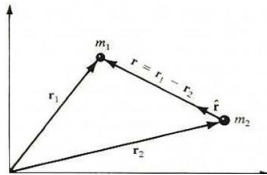
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3-D rotational motion :
particle on a sphere



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rigid rotor : $|\vec{r}|$ fixed, only rotational motion of coordinate $\vec{r}$ observed



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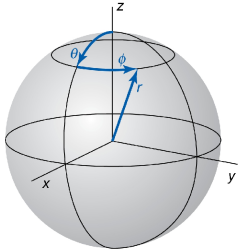
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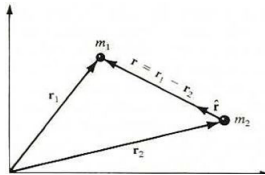
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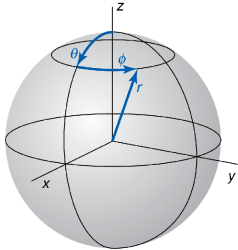
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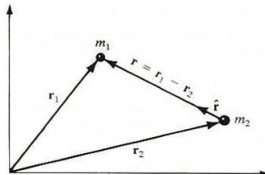
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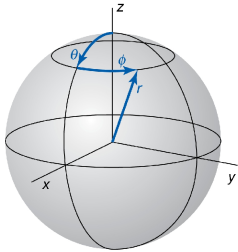
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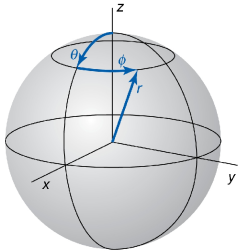
Angular momentum algebra:

Converting partial derivatives : Cartesian to spherical polar



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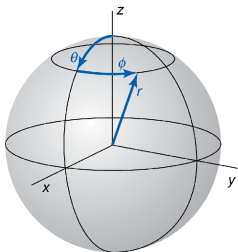
Converting partial derivatives : Cartesian to spherical polar



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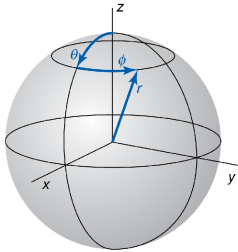
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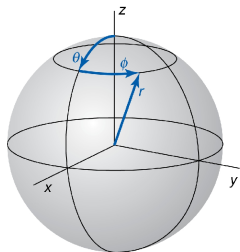
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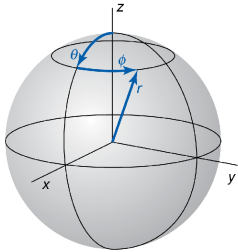
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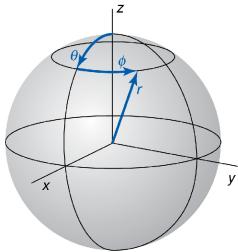
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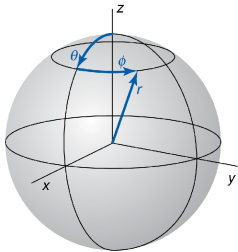
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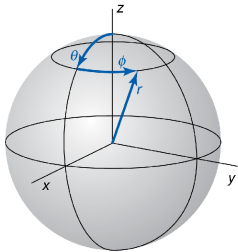
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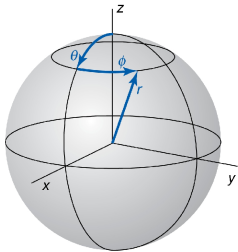
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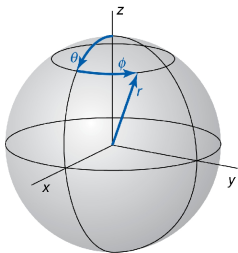
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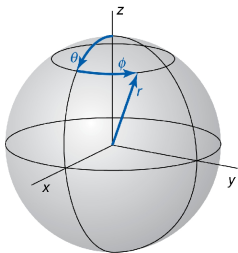
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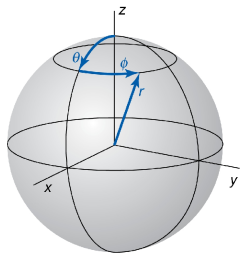
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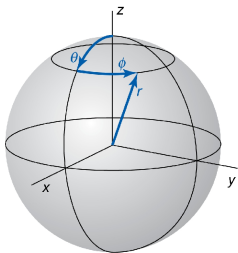
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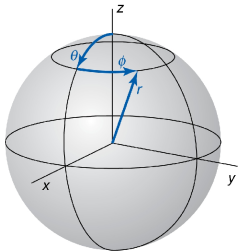
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$\therefore z$ has no ϕ dependence, $c_3 = 0$

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$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 = \frac{1}{2} (\hat{L}_+ \hat{L}_- + \hat{L}_- \hat{L}_+) + \hat{L}_z^2$$

express components of angular momenta in spherical polar coordinates:

$$\frac{i}{\hbar} \hat{L}_x = \frac{i}{\hbar} (\hat{y} \hat{p}_z - \hat{z} \hat{p}_y) = y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y}$$

$$\begin{pmatrix} \hat{L}_x \\ \hat{L}_y \end{pmatrix} = \frac{\hbar}{i} \begin{pmatrix} -\sin \phi & -\cos \phi \\ \cos \phi & \sin \phi \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial \theta} \\ \cot \theta \frac{\partial}{\partial \phi} \end{pmatrix} \text{ and } \hat{L}_z = \frac{\hbar}{i} \frac{\partial}{\partial \phi}$$

no r -dependence (property independent of radial distance from centre of rotation)

$$\text{define } \hat{L}_{\pm} = \hat{L}_x \pm i \hat{L}_y = \frac{\hbar}{i} \left[(\sin \phi \pm i \cos \phi) \frac{\partial}{\partial \theta} - \cot \theta (\cos \phi \pm i \sin \phi) \frac{\partial}{\partial \phi} \right]$$

$$\hat{L}_{\pm} = \pm \hbar e^{\pm i \phi} \left(\frac{\partial}{\partial \theta} \pm i \cot \theta \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 = \frac{1}{2} (\hat{L}_+ \hat{L}_- + \hat{L}_- \hat{L}_+) + \hat{L}_z^2$$

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] = -\hbar^2 \Lambda^2$$

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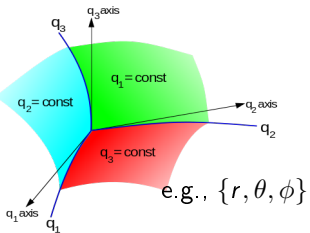
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operator Λ^2 : Legendrian

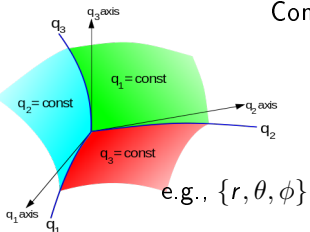
mathematical digression -

mathematical digression - curvilinear coordinates q_1, q_2, q_3



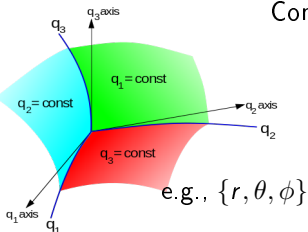
mathematical digression - curvilinear coordinates q_1, q_2, q_3

Consider differential increments $ds_i \perp$ surface $q_i = \text{constant}$

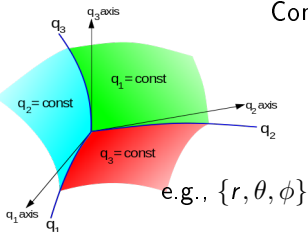


mathematical digression - curvilinear coordinates q_1, q_2, q_3

Consider differential increments $ds_i \perp$ surface $q_i = \text{constant}$
increments are related to dq_i as $ds_i = h_i \cdot dq_i$



mathematical digression - curvilinear coordinates q_1, q_2, q_3

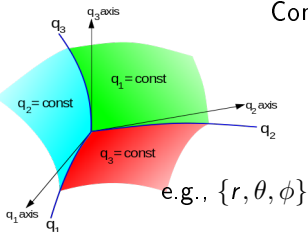


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mathematical digression - curvilinear coordinates q_1, q_2, q_3

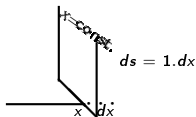


Consider differential increments $ds_i \perp$ surface $q_i = \text{constant}$

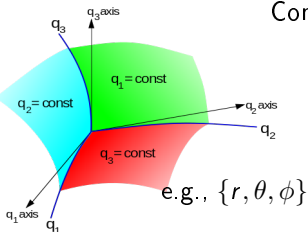
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Cartesian $\therefore ds_i = 1 \cdot dx_i; \therefore h_i = 1 \quad \forall i = 1, 2, 3$



mathematical digression - curvilinear coordinates q_1, q_2, q_3

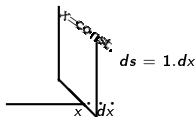


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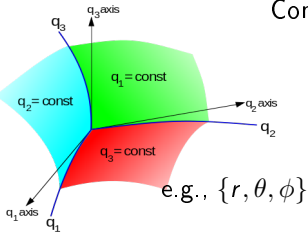
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$$\vec{\nabla} f = \hat{x} \frac{\partial f}{\partial x} + \hat{y} \frac{\partial f}{\partial y} + \hat{z} \frac{\partial f}{\partial z}$$

mathematical digression - curvilinear coordinates q_1, q_2, q_3

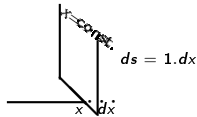


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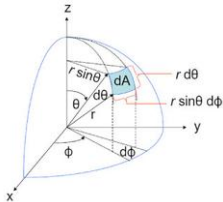
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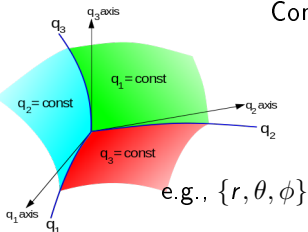


$$\vec{\nabla} f = \hat{x} \frac{\partial f}{\partial x} + \hat{y} \frac{\partial f}{\partial y} + \hat{z} \frac{\partial f}{\partial z}$$



spherical polar

mathematical digression - curvilinear coordinates q_1, q_2, q_3

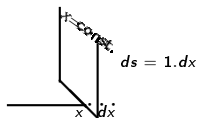


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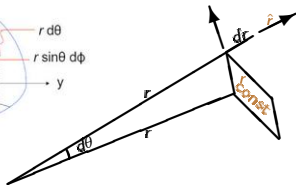
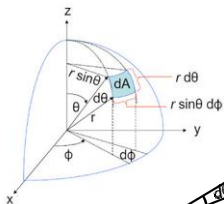
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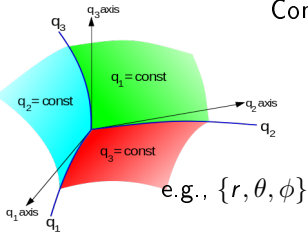


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spherical polar

mathematical digression - curvilinear coordinates q_1, q_2, q_3

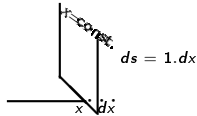


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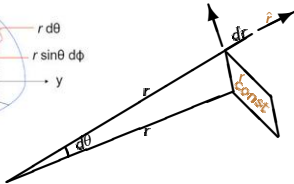
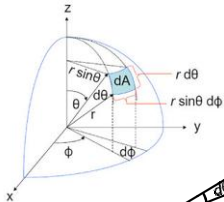
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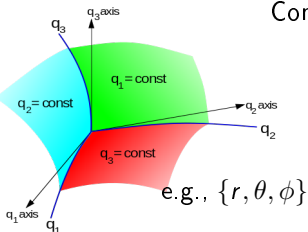
$$\vec{\nabla} f = \hat{x} \frac{\partial f}{\partial x} + \hat{y} \frac{\partial f}{\partial y} + \hat{z} \frac{\partial f}{\partial z}$$



$$ds_1 = 1 \cdot dr$$

spherical polar

mathematical digression - curvilinear coordinates q_1, q_2, q_3

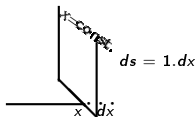


Consider differential increments $ds_i \perp$ surface $q_i = \text{constant}$

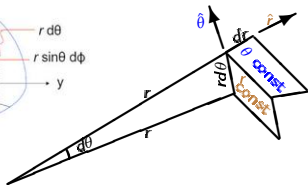
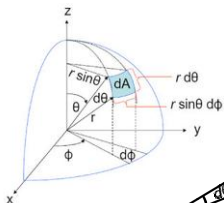
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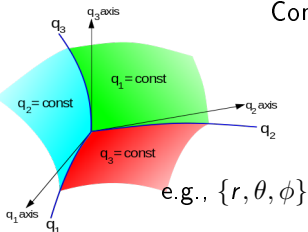


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spherical polar

mathematical digression - curvilinear coordinates q_1, q_2, q_3

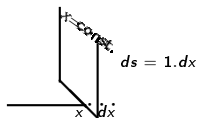


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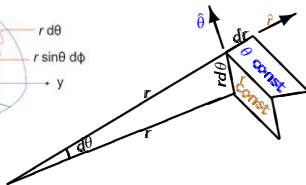
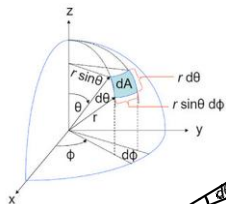
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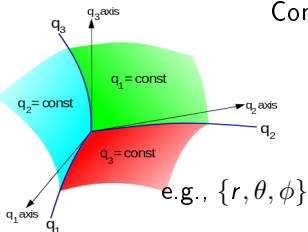
$$\vec{\nabla} f = \hat{x} \frac{\partial f}{\partial x} + \hat{y} \frac{\partial f}{\partial y} + \hat{z} \frac{\partial f}{\partial z}$$



$$ds_1 = 1 \cdot dr; ds_2 = r d\theta;$$

spherical polar

mathematical digression - curvilinear coordinates q_1, q_2, q_3

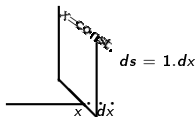


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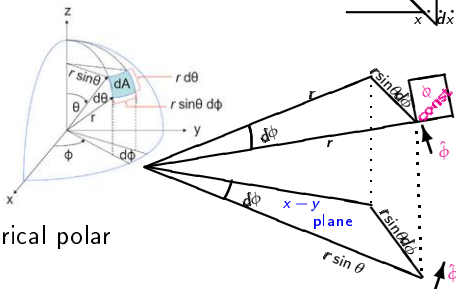
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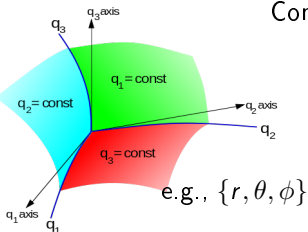


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spherical polar

mathematical digression - curvilinear coordinates q_1, q_2, q_3

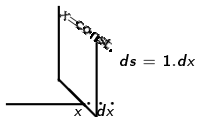


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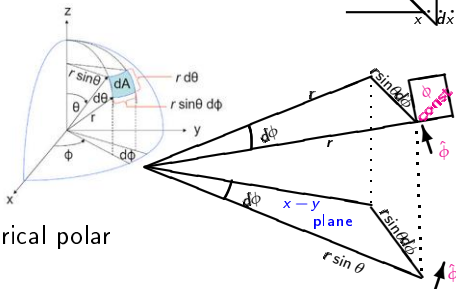
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$$ds_1 = 1 \cdot dr; ds_2 = r d\theta; ds_3 = r \sin \theta d\phi$$

spherical polar

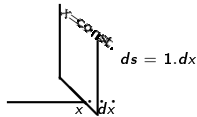
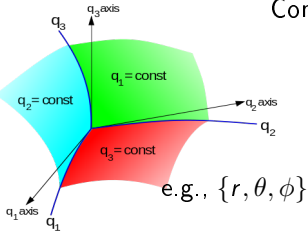
mathematical digression - curvilinear coordinates q_1, q_2, q_3

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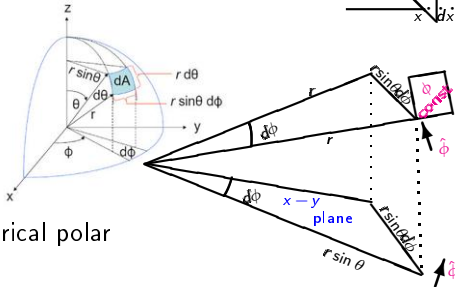
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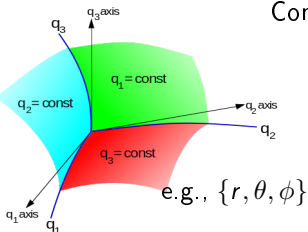


$$ds_1=1.dr; ds_2=r d\theta; ds_3=r \sin \theta d\phi$$

$$\therefore h_1 = 1; h_2 = r; h_3 = r \sin \theta$$

spherical polar

mathematical digression - curvilinear coordinates q_1, q_2, q_3

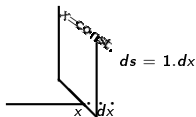


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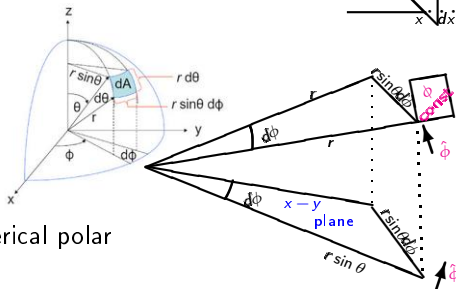
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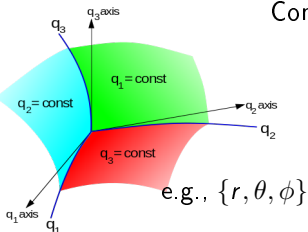
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mathematical digression - curvilinear coordinates q_1, q_2, q_3

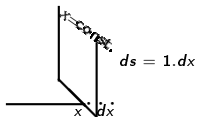


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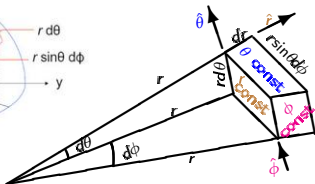
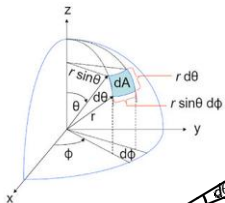
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Cartesian $\therefore ds_i = 1 \cdot dx_i; \therefore h_i = 1 \quad \forall i = 1, 2, 3$



$$\vec{\nabla} f = \hat{x} \frac{\partial f}{\partial x} + \hat{y} \frac{\partial f}{\partial y} + \hat{z} \frac{\partial f}{\partial z}$$



$$ds_1 = 1 \cdot dr; ds_2 = r d\theta; ds_3 = r \sin \theta d\phi$$

$$\therefore h_1 = 1; h_2 = r; h_3 = r \sin \theta$$

$$\vec{\nabla} f = \hat{r} \frac{\partial f}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial f}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi}$$

spherical polar

$$\text{Divergence } \vec{\nabla} \cdot \vec{V} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 V_1) + \frac{\partial}{\partial q_2} (h_3 h_1 V_2) + \frac{\partial}{\partial q_3} (h_1 h_2 V_3) \right]$$

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This equation has meaningful, well-behaved solutions iff

$$l = |m|, |m-1|, |m-2|, \dots$$

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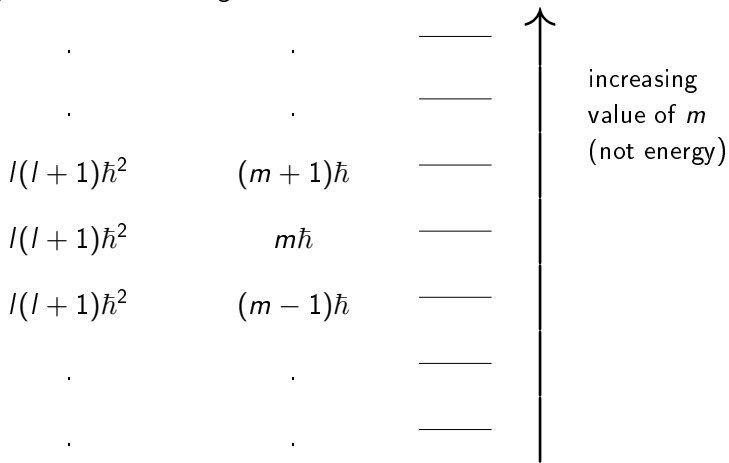
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it is a simultaneous eigenfunction of $\hat{L}^2$ and $\hat{L}_z$, with eigenvalues $l(l+1)\hbar^2$ for the operator $\hat{L}^2$ and $(m+1)\hbar$ for the operator $\hat{L}_z$

basis set $\{|l, m\rangle\}$ arranged according to the m -value.

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But this cannot go forever

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obtain all other $Y_l^m(\Omega)$ by using the ladder operator

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$$\left(\frac{\hat{L}_+}{\hbar}\right)^k |l, -l\rangle = \sqrt{2l(2l-1)(2l-2)\cdots(2l+1-k)} \sqrt{1.2.\cdots k} |l, -l+k\rangle$$

In particular, choose $k = l + m$,

$$\begin{aligned} \left(\frac{\hat{L}_+}{\hbar}\right)^{l+m} |l, -l\rangle &= \sqrt{2l.1(2l-1)(2l-2)\cdots(l+1-m)} \sqrt{1.2.\cdots(l+m)} |l, m\rangle \\ &= \sqrt{\frac{2l!(l+m)!}{(l-m)!}} |l, m\rangle \end{aligned}$$

Using $\frac{\hat{L}_+}{\hbar} |lm\rangle = \sqrt{(l+m+1)(l-m)} |l, m+1\rangle$, we have

$$\frac{\hat{L}_+}{\hbar} |l, -l\rangle = \sqrt{2l.1} |l, -l+1\rangle$$

$$\left(\frac{\hat{L}_+}{\hbar}\right)^2 |l, -l\rangle = \sqrt{2l.1} \sqrt{(2l-1).2} |l, -l+2\rangle$$

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$$\therefore |l, m\rangle = \sqrt{\frac{(l-m)!}{2l!(l+m)!}} \left(\frac{\hat{L}_+}{\hbar}\right)^{l+m} |l, -l\rangle$$

Recall : $\frac{\hat{L}_+}{\hbar} = e^{i\phi} \left(\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right)$

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$$Y_l^m(\Omega) = (-1)^{l+m} \left[\sqrt{\frac{(2l+1)}{4\pi}} \sqrt{\frac{(l-m)!}{(l+m)!}} \cdot \frac{1}{2^l l!} \sin^m \theta \cdot \frac{d^{l+m}}{d(\cos \theta)^{l+m}} \sin^{2l} \theta \right] e^{im\phi}$$

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Define $P_l(\zeta) = \frac{1}{2^l l!} \cdot \frac{d^l}{d\zeta^l} (\zeta^2 - 1)^l$; here $\zeta = \cos \theta$: Legendre polynomial

$$\frac{\hat{L}_\pm}{\hbar} [F(\theta) e^{i\nu\phi}] = (-1)^1 e^{i(\nu+1)\phi} \sin^{\nu+1} \theta \frac{\partial}{\partial(\cos \theta)} \left(\frac{F(\theta)}{\sin^\nu \theta} \right)$$

take $F_1(\theta) = -\sin^{\nu+1} \theta \frac{\partial}{\partial(\cos \theta)} \left(\frac{F(\theta)}{\sin^\nu \theta} \right)$ and change ν to $\nu + 1$ on LHS,

$$\begin{aligned} \text{then } \frac{\hat{L}_\pm}{\hbar} [F_1(\theta) e^{i(\nu+1)\phi}] &= (-1)^1 e^{i(\nu+2)\phi} \sin^{\nu+2} \theta \frac{\partial}{\partial(\cos \theta)} \left(\frac{F_1(\theta)}{\sin^{\nu+1} \theta} \right) \\ &= (-1)^2 e^{i(\nu+2)\phi} \sin^{\nu+2} \theta \frac{\partial}{\partial(\cos \theta)} \left(\frac{\cancel{\sin^{\nu+1} \theta} \frac{\partial}{\partial(\cos \theta)} \left(\frac{F(\theta)}{\sin^\nu \theta} \right)}{\cancel{\sin^{\nu+1} \theta}} \right) \\ &= (-1)^2 e^{i(\nu+2)\phi} \sin^{\nu+2} \theta \frac{\partial^2}{\partial(\cos \theta)^2} \left(\frac{F(\theta)}{\sin^\nu \theta} \right) \end{aligned}$$

$$\text{Proceeding } n \text{ times: } \left(\frac{\hat{L}_\pm}{\hbar} \right)^n [F(\theta) \cdot e^{i\nu\phi}] = (-1)^n e^{i(\nu+n)\phi} \sin^{\nu+n} \theta \frac{\partial^n}{\partial(\cos \theta)^n} \left(\frac{F(\theta)}{\sin^\nu \theta} \right)$$

Choose $n = l + m$, $\nu = -l$ and $Y_l^{-l}(\Omega) = F(\theta) e^{-il\phi}$, where $F(\theta) = \frac{1}{2^l l!} \sqrt{\frac{(2l+1)!}{4\pi}} \sin^l \theta$

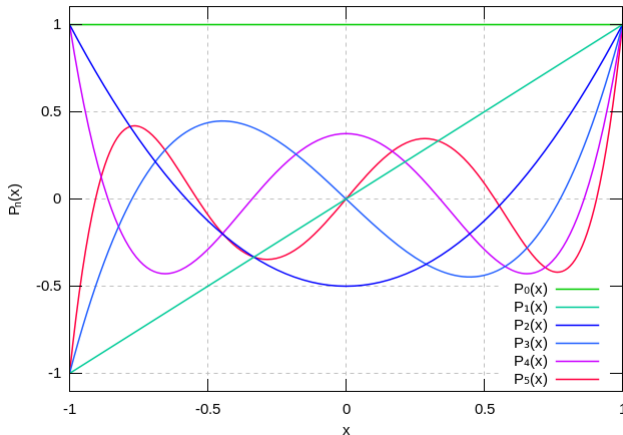
$$\begin{aligned} Y_l^m(\Omega) &= (-1)^{l+m} \left[\sqrt{\frac{(2l+1)}{4\pi}} \sqrt{\frac{(l-m)!}{(l+m)!}} \cdot \frac{1}{2^l l!} \sin^m \theta \cdot \frac{d^{l+m}}{d(\cos \theta)^{l+m}} \sin^{2l} \theta \right] e^{im\phi} \\ &= (-1)^m \sqrt{\frac{(2l+1)}{4\pi}} \sqrt{\frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{im\phi} \end{aligned}$$

Define $P_l(\zeta) = \frac{1}{2^l l!} \cdot \frac{d^l}{d\zeta^l} (\zeta^2 - 1)^l$; here $\zeta = \cos \theta$: Legendre polynomial

$P_l^m(\theta) = (-1)^m \cdot (1 - \zeta^2)^{\frac{m}{2}} \cdot \frac{d^m}{d\zeta^m} P_l(\zeta)$: associated Legendre polynomial

$P_l(x)$ is a l -th degree polynomial
(has l nodes)

legendre polynomials



associated Legendre
polynomials

$$P_0^0(\cos \theta) = 1$$

$$P_1^0(\cos \theta) = \cos \theta$$

$$P_1^1(\cos \theta) = -\sin \theta$$

$$P_2^0(\cos \theta) = \frac{1}{2}(3\cos^2 \theta - 1)$$

$$P_2^1(\cos \theta) = -3\cos \theta \sin \theta$$

$$P_2^2(\cos \theta) = 3\sin^2 \theta$$

$$P_3^0(\cos \theta) = \frac{1}{2}(5\cos^3 \theta - 3\cos \theta)$$

$$P_3^1(\cos \theta) = -\frac{3}{2}(5\cos^2 \theta - 1)\sin \theta$$

$$P_3^2(\cos \theta) = 15\cos \theta \sin^2 \theta$$

$$P_3^3(\cos \theta) = -15\sin^3 \theta$$

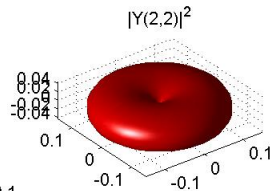
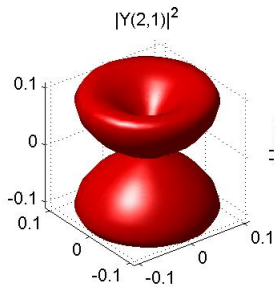
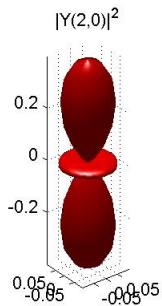
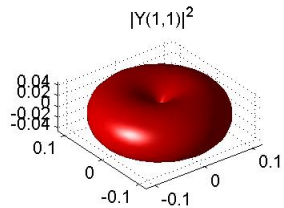
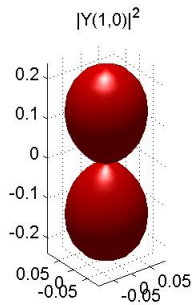
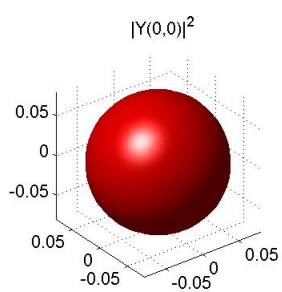
$$P_4^0(\cos \theta) = \frac{1}{8}(35\cos^4 \theta - 30\cos^2 \theta + 3)$$

$$P_4^1(\cos \theta) = -\frac{5}{2}(7\cos^3 \theta - 3\cos \theta)\sin \theta$$

$$P_4^2(\cos \theta) = \frac{15}{2}(7\cos^2 \theta - 1)\sin^2 \theta$$

$$P_4^3(\cos \theta) = -105\cos \theta \sin^3 \theta$$

$$P_4^4(\cos \theta) = 105\sin^4 \theta$$



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